

CHARLTON, Australia

WMO: 948390

Lat: 36.2847S

Lon: 143.3342E

Elev: 434

StdP: 14.47

Time Zone: 10.00 (AUS)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	31.9	34.0	22.8	17.5	78.0	26.6	20.9	73.1	20.2	52.7	18.2	53.4	1.1	190	0.491

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	28.9	102.1	65.9	97.4	64.7	93.4	64.0	70.6	85.3	68.8	84.8	67.1	85.1	10.5	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.1	101.3	73.8	64.3	91.8	73.3	61.8	83.8	71.5	34.8	82.9	33.2	84.5	31.8	85.1	77.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 18.9	16.7	15.1	DB	27.8	109.5	2.5	2.8	26.1	111.5	24.6	113.1	23.3	114.6	21.5	116.7	(4)
(5)			WB	27.7	73.5	2.6	2.3	25.8	75.1	24.3	76.5	22.8	77.8	20.9	79.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	60.3	74.6	72.9	68.4	60.9	53.8	48.7	47.8	49.1	53.0	59.4	65.7	69.9
	DBStd	11.35	7.90	7.34	7.46	6.21	4.81	3.67	3.80	4.20	5.48	7.21	8.10	7.78
	HDD50	265	0	0	0	2	14	65	89	68	23	4	0	0
	HDD65	2773	7	13	46	153	348	488	533	493	364	208	87	33
	CDD50	4021	763	642	571	330	132	28	21	40	112	295	471	616
	CDD65	1054	305	235	152	31	1	0	0	0	3	35	108	184
	CDH74	15005	4425	3045	1866	393	6	0	0	3	64	671	1709	2823
	CDH80	7516	2499	1548	840	103	0	0	0	0	12	252	816	1446
Wind	WSAvg	7.3	8.5	8.3	7.4	6.2	6.4	5.7	6.7	6.8	7.2	7.7	8.2	8.5
Precipitation	PrecAvg	14.5	0.8	0.8	0.7	0.8	1.2	1.8	1.6	1.4	1.8	1.1	1.3	1.1
	PrecMax	24.8	5.9	2.8	2.8	2.9	3.1	4.6	3.5	3.8	5.7	4.3	4.5	4.6
	PrecMin	8.0	0.0	0.0	0.0	0.0	0.0	0.6	0.3	0.0	0.6	0.0	0.0	0.0
	PrecStd	4.5	1.2	0.8	0.6	0.8	0.8	1.0	0.7	0.9	1.2	1.1	1.1	1.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	108.9	105.5	100.0	88.9	74.7	63.3	62.9	71.4	81.2	93.1	99.6	105.2
		MCWB	67.7	68.1	65.0	62.1	57.5	53.8	52.2	54.5	56.4	60.3	63.5	65.1
	2%	DB	103.7	97.9	94.1	82.5	69.5	60.6	59.7	65.4	74.7	86.8	93.8	98.8
		MCWB	66.6	66.3	64.6	60.3	56.0	52.4	52.6	52.8	56.1	59.2	62.6	64.1
	5%	DB	97.8	93.5	88.9	78.5	66.1	58.6	57.3	61.6	69.7	81.5	89.3	93.3
		MCWB	65.3	65.2	63.7	58.7	55.4	51.9	51.0	51.9	55.1	57.7	61.9	62.7
	10%	DB	92.8	89.5	84.2	74.4	63.3	56.8	55.3	58.4	65.2	76.4	84.2	88.3
		MCWB	64.4	63.8	62.4	57.6	54.4	51.2	50.2	50.8	53.0	56.8	60.5	61.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	73.6	72.9	69.7	64.5	60.8	56.8	56.2	57.0	60.9	65.4	68.9	71.0
		MCDB	87.2	87.1	81.6	75.8	65.5	58.5	59.4	65.0	71.0	78.6	83.3	84.1
	2%	WB	70.9	70.5	67.6	62.8	58.5	54.7	53.8	54.7	58.2	62.3	66.6	68.0
		MCDB	87.4	85.0	83.3	73.3	65.5	57.9	57.1	62.5	69.3	76.3	82.2	82.6
	5%	WB	68.9	68.3	65.9	61.0	57.0	53.4	52.2	52.9	56.1	60.1	64.4	65.8
		MCDB	87.2	83.9	80.5	73.2	64.1	56.8	55.8	59.6	66.7	75.1	81.7	85.2
	10%	WB	66.8	66.4	64.2	59.3	55.5	52.2	50.8	51.4	54.2	58.1	62.4	63.9
		MCDB	88.5	85.1	80.6	72.4	61.6	55.7	54.3	57.3	63.8	73.7	78.8	83.6
Mean Daily Temperature Range	5% DB	MDBR	28.9	27.4	24.9	23.6	19.0	17.0	15.8	18.8	22.9	27.6	27.8	28.7
		MCDBR	35.8	32.7	30.6	29.3	22.4	19.5	17.8	24.1	29.1	35.6	34.8	35.5
		MCWBR	10.9	10.4	10.2	12.2	11.9	12.4	11.4	13.4	13.8	13.9	11.1	11.4
	5% WB	MCDBR	26.7	25.5	23.7	23.1	18.2	16.2	14.7	20.3	24.1	28.0	27.0	25.8
		MCWBR	10.5	10.1	9.4	10.5	10.3	11.3	10.2	12.4	13.1	12.9	10.1	10.6
Clear-Sky Solar Irradiance	taub	0.335	0.332	0.318	0.307	0.297	0.288	0.291	0.299	0.323	0.329	0.338	0.328	
	taud	2.503	2.523	2.552	2.562	2.572	2.594	2.580	2.539	2.457	2.453	2.464	2.514	
	Ebn at Noon	317	311	304	288	273	267	273	287	296	308	314	321	
	Edh at Noon	36	34	30	26	23	21	22	26	33	36	37	36	
All-Sky Solar Radiation	RadAvg	2477	2213	1777	1279	869	685	735	1037	1442	1917	2218	2468	
	RadStd	118	106	83	60	44	58	40	72	102	133	129	108	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (23 neighbors)	+0.85	N/A	N/A	+1.57	N/A	N/A	N/A	-158	+272	+146

Nomenclature: See separate page